SEAGLASS FOUNDRY

Forged For Speed. Refined by Craft.

ForgeGIS Studio™

Where You See the Forge Suite Work — and Shape It to Your Mission

ForgeGIS™ computes, ForgeData™ catalogs and ForgeMind™ reasons — but each speaks only libraries and machine protocols. ForgeGIS Studio™ is the browser workspace that puts them in front of a person: a live map, a pipeline canvas, a batch runner, and a natural-language agent that acts on the map you are looking at. It is not a general-purpose GIS desktop; it is the fastest way to see what the suite does, and a thin surface built to be re-cut to your workflow.

1 host

the whole suite on one workstation

Studio binds to loopback and is reached only from that machine's browser

3 engines

behind one workspace

GPU compute, the data catalog and the agent, operated from the same map

R · V · M

raster, vector and mixed pipelines

linear chains run as you build them; branching graphs run in the batch runner

What it is

A browser workspace, backed by a local server, that loads data through one catalog, composes raster, vector and mixed pipelines on a node canvas, runs them across a folder in batch, and hosts the agent conversation. Operations arrive from the ForgeGIS catalog and appear as nodes automatically.

New in 1.2.6

Raster batches run on the suite's compute broker when one is installed, and fall back to this machine when it cannot take them. The rest is about being right: elevations print in their own vertical units, never an assumed metre; measurements are on the WGS84 ellipsoid; and a raster the map cannot place is flagged or refused, not drawn in the wrong spot.

Why it matters

The engines are deliberately headless, which keeps each one focused — but a person needs somewhere to stand. Studio is that place: the component whose job is to make the suite visible and usable, so an evaluator sees GPU pipelines run, graphs branch and an agent act on a real map in the first session.

In the suite. Studio is an opt-in component of the Forge suite, installed and started by the Seaglass Control Center™ and opened in the browser from its dashboard. It runs on one Windows x64 workstation, bound to loopback; when the Control Center also installs the suite's compute broker, it points Studio's raster batches at it. The basemap and URL layers need a network, while rasters render offline.

Checked on screen. Release 1.2.6 closed with a testing audit and an end-to-end suite that drives Studio in a real browser against the real engine and checks the value under each view against GDAL. Every case was first shown to fail on a planted fault, and Measure's lengths and areas are pinned to GeographicLib and pyproj values.

CONTINUED ON REVERSE →

WHAT AN OPERATOR DOES IN STUDIO

Compose pipelines by hand

Operations become nodes; connecting them builds a pipeline that can cross from raster into vector — a Polygonize → Buffer chain is drawn directly, and the palette only offers operations that fit the node you are on. Graphs save to a library in one versioned format; linear chains replay on the canvas, and any saved graph runs in batch; pipelines converted from QGIS models carry their import warnings with them.

Run one, then run hundreds

The batch runner applies a saved pipeline to a folder of inputs. Raster batches go to the suite's compute broker when one is installed; the rest run on this machine, which checks each tile's GPU memory estimate as it is dispatched so a tile that would not fit is recorded rather than crashing the run. A dry run shows what will happen first; every run ends with a reason and a report.

Let the agent drive the map

Ask in Studio's chat and ForgeMind can act on the canvas — place named markers, draw overlays, move the camera, push reports — through an authenticated bridge. Named results are kept in the ForgeData catalog, so they survive a restart with the same identities.

> **Built to be tailored.** Studio is thin over stable engines. A ForgeGIS operation of a kind Studio already carries — raster, vector, or raster ↔ vector — is authorable, runnable and batchable with no Studio code change; workflows are saved data, not code; and extending what the agent can do on the map is adding a tool to the bridge. Shaping Studio to a customer's task means composing known pieces, not building a platform.

> **Hardened in 1.2.6.** Studio refuses any request not addressed to a loopback host, which closes off DNS rebinding from a web page; Tools-panel calls are checked exactly as the ForgeGIS MCP server checks them, so a misspelled argument is refused rather than ignored; and the agent bridge's tools are pinned by a versioned contract.

Get the Technical Brief

The Technical Brief covers the architecture, the canvas and batch runner, the agent bridge, data access, security posture and deployment. ForgeGIS Studio is part of the Forge suite and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified teams.

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